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DATABASE SYSTEM

DATABASE PROJECT

FITNESS CLUB MEMBERSHIP

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Table of Content

1.0	Introduction	
1.1	Project Overview	3
1.2	Objectives of the Database Project	4
1.3	Main Entities and Relationships	6
2.0	Database Design	
2.1	Entity-Relationship Diagram (ERD)	12
2.2	Mapping to Relational Schema	14
2.3	Data Dictionary	15
3.0	Database Implementation	
3.1	SQL Code for Database Creation	26
3.2	Database Population	31
3.3	Join Data	38
3.4	SQL Queries	46
3.5	Update and Delete Queries	52
4.0	Conclusion	56
5.0	Appendix	57

1.0 Introduction

1.1 Project Overview

The Fitness Club Membership System is a database system that are designed to make the operations of a fitness club more efficient by overseeing member details, class schedules, trainers, payments, and equipment maintenance. The primary goal of this system is to improve the effectiveness of membership management, provide clear tracking of member activity, and support the club administrator in organizing classes and maintaining equipment. It is also allowing club management to track and manage all essential aspects of the club's day-to-day activities for smooth operations.

One of the main goals of this system is to streamline member information management by consolidating data that is currently dispersed among different manual processes or separate systems. This will enable the system to track member activities more accurately and in real-time, including class attendance, participation in fitness programs, and use of club facilities. The system not only monitors member activity but also provides club management with a user-friendly platform to oversee class schedules and instructor assignments, ensuring resources are allocated effectively to meet member needs.

Then the system ability is to process payments efficiently, whether for membership renewals, class fees, or personal training sessions. By automating these financial transactions, the system not only can improve the speed and accuracy of payments, but it is also reducing the administrative burden on staff, that can make them be more focus on enhancing the member experience. Additionally, the system can also generate detailed financial reports that can ease the management to monitor revenue streams, analyse cash flow, and forecast future growth opportunities.

Despite of that, the system will also focus on maintaining equipment. The system is responsible in tracking usage patterns and make the schedule of regular maintenance to ensure that the club's facilities are always in optimal condition. This feature will reduce the likelihood of equipment breakdowns, thus minimize the disruptions to members and avoiding costly repairs. The system will automatically alert staff when maintenance is due, ensuring timely upkeep of all equipment and facilities.

Another key aspect, efficiently achieved without any privacy concerns, is data security and integrity. When managing a fitness club, personal and financial data is very sensitive in nature so the system will use several measures to secure member data from getting into the wrong hands and include encryption, role-based access control, and regular backups. The system will also follow applicable data protection laws, meaning that members' information is always treated in a most private and confidential manner.

In summary, the Fitness Club Membership System is a versatile and scalable database solution that will revolutionize how fitness clubs manage their operations. By automating membership tracking, class scheduling, payments, and equipment maintenance, the system will reduce administrative workloads, improve member satisfaction, and provide valuable insights for business decision-making. With its ability to adapt to future growth and ensure data security, this system will serve as a long-term asset for any fitness club looking to enhance operational efficiency and provide a superior member experience.

1.2 Objectives

The main objective of this project is to develop a robust and efficient database system that supports the management of fitness club memberships, programs, and staff, allowing for better decision-making and improved operations within the club. The system will serve as a unified platform for processing and storing critical information about students, instructors, staff, and equipment, ensuring that all relevant information is current and available. Efficient management of member information including joining date, membership type, payment history and renewal of memberships. This integration will be facilitated in addition to tracking the participation of each member in the exercise process. To helps clubs better understand the needs of their members and improve class offerings.

Role-based access controls are integrated into the system to protect the privacy and security of club data. This ensures that sensitive information such as employee salaries or payment details is captured. It will only be accessible by authorized persons. This will help ensure compliance with relevant data protection laws and protects data integrity. Database systems automate many important processes including the billing cycles, class attendance tracking, and registration of new members. By automating these processes will reduce the chance of human errors, administrative burdens, and increase overall accuracy and efficiency.

Additionally, the system includes a detailed reporting module that can produce in-depth reports on a range of club operations. Vital information about member growth patterns, class popularity, peak usage period and overall financial success is passed on to management. Club management will be better equipped to allocate resources, optimize the class schedule, make an educated decision, and modify pricing structures with the use of these insights.

The system is also designed to work with popular fitness trackers and mobile apps. It allows members to link their fitness progress to their club profile. This integration will enable members to track their workouts, view their progress and receive personalized fitness recommendations based on their activity. Real time data on equipment usage and staff availability helps clubs better manage their resources. This ensures that the exercise equipment is properly maintained and there are enough staff during peak hours to provide the best service to members.

Scalability is important consideration in database system design. As fitness clubs grow either by increasing the members base, opening a new branch, or expanding the fitness club program, the system will be able to accommodate this growth without the need for restructuring or additional infrastructure. This flexibility helps ensure that the database evolves over time. This flexibility will ensure the database remains a valuable resource for the club and the system will increase operational efficiencies as well as improve member experience and support the long-term success of the club.

1.3 Main entities and Relationship:

The main entities included:

- **Member:**

Represents individuals who have joined the fitness club. Each member has a unique profile that includes personal details (name, contact information), membership type, start and end dates, and address. A member may register for multiple fitness classes and can hold one or more membership plans.

- **Trainer:**

Represents staff members that responsible for conducting fitness classes and providing personal training sessions. Trainers are assigned to one or more fitness programs or classes and interact with the members during those sessions.

- **Class:**

Represents fitness sessions offered by the club. Each class has a schedule, a maximum number of participants, and is led by a trainer. Members can register for classes, and attendance is tracked for reporting purposes.

- **Payment:**

Represents financial transactions made by members to the fitness club for services like membership renewal, class registration, and other club services. Each payment is linked to a membership plan and includes details like payment date, amount, and method of payment.

- **Equipment:**

Represents the various fitness equipment available for use at the club. The system tracks each piece of equipment, including its purchase date, maintenance schedule, and status. This helps ensure that the equipment is well-maintained and available for member use.

- **Membership Plan**

Represents the different plans offered by the fitness club. Each membership plan has details like plan type (monthly, yearly, etc.), cost, benefits, and associated access to facilities or classes. A member can subscribe to one or more membership plans.

- **Facility**
Represents to the various amenities available at the fitness club, such as gyms, swimming pools, saunas, or tennis courts. Each facility has attributes like location, capacity, and availability. Facilities can be associated with specific classes or membership types.
- **Attendance**
Tracks the presence of members at the club. Attendance records include member ID, date and time of entry, and facility or class accessed. It helps monitor usage patterns and enforce club rules regarding attendance limits or class capacity.
- **Workout Plan**
Represents personalized or standard exercise plans for members. Each workout plan is created by trainers based on a member's fitness goals, health records, and preferences. It contains details like exercise routines, frequency, and duration.
- **Trainer Certification**
Contains information about the qualifications and certifications held by trainers. It includes certification type, issuing authority, expiration date, and renewal status. Trainers may have multiple certifications that qualify them to lead specific classes or offer specialized training.
- **Health Record**
Stores health-related information about members, such as medical conditions, fitness assessments, and dietary preferences. This data ensures personalized workout plans and minimizes health risks during physical activities.
- **Feedback**
Captures feedback from members about the club's services, trainers, facilities, and overall experience. It includes member ID, trainer or facility ID, rating, and comments. This information is valuable for improving the quality of services.
- **Subscription**
Refers to the ongoing commitment a member makes to a particular membership plan or service. Subscriptions include start and end dates, payment frequency, and any associated facilities or classes that the member has access to.

- **Promotion**
Represents special offers or discounts provided to attract new members or retain existing ones. Promotions have attributes such as discount type, eligibility criteria, duration, and services covered (e.g., discounts on specific membership plans or classes).
- **Session**
Refers to a scheduled period of time during which a fitness class or personal training is conducted. Sessions are linked to trainers, members, and specific classes, and include details like time, location, and maximum number of participants.
- **Invoice**
Represents financial transactions between the member and the fitness club. Invoices include payment details for membership plans, classes, or additional services, with fields like date, amount, payment method, and due date
- **Locker**
Represents the lockers assigned to members for storing personal belongings. Each locker is identified by a unique number, and the assignment includes information such as locker size, duration of use, and associated fees (if applicable).
- **Referral**
Tracks member referrals, where existing members refer new individuals to join the club. Referrals may be tied to promotional offers, and the system records details about both the referring and referred members.
- **Inventory**
Manages the stock of equipment, supplements, or other goods available at the club. Inventory includes item names, quantities, supplier details, and restocking schedules, ensuring the club has the necessary resources for members and trainers.
- **Trainer schedule**
Represents the availability of trainers and the classes they lead. Each trainer's schedule includes session times, assigned facilities, and any one-on-one training appointments, allowing for effective class management and trainer allocation.

The relationships include:

Member-Class: A many-to-many relationship where members can enrolment in multiple classes. Each member can enrol in multiple classes, and each class can have multiple members.

Member-Trainer: A many-to-many relationship tracks personal training relationships between members and trainers. Members may work with multiple trainers, and trainers can coach multiple members.

Member-Session: A many-to-many relationship where members can attend multiple sessions, and each session can have multiple members attending.

Member-Payment: A one-to-many relationship tracking payments for each member. Each member can make multiple payments, whether for memberships or classes.

Member-Equipment: A many-to-many relationship tracking the equipment used by the members. Members use different equipment during workouts, and each piece of equipment can be used by multiple members.

Member-Membership Plan: A many-to-many relationship where each member can subscribe to multiple membership plans, and each membership plan can be associated with multiple members.

Member-Attendance: A one-to-many relationship where each member has multiple attendance records. Each attendance record tracks the member's presence at a class or facility at a specific time.

Member-Workout Plan: A one-to-many relationship where each member can follow multiple workout plans. Workout plans are created based on the member's fitness goals, and a member may switch between plans over time.

Member-Health Record: A one-to-one relationship where each member has one health record, which contains essential health information used to personalize their workout plans.

Member-Feedback: A one-to-many relationship where members can provide feedback on various aspects of the fitness club. Each member may leave multiple feedback entries on trainers, facilities, or classes.

Member-Subscription: A one-to-many relationship where members have subscriptions to various services or membership plans. Each subscription represents the member's commitment to a specific service or plan over time.

Member-Referral: A one-to-many relationship where a member may refer multiple individuals to the fitness club. Each referral links one member with a referred individual.

Member-Locker:

A one-to-one relationship where each member can be assigned a locker for storing personal belongings during their time at the club.

Member-Invoice:

A one-to-many relationship where each member generates multiple invoices for services, such as membership fees or class registrations.

Trainer-Equipment: A one-to-many relationship tracks the equipment assigned to trainers for classes. Trainers use multiple pieces of equipment for their classes or personal training sessions, but each piece can only be assigned to one trainer at a time to avoid scheduling conflicts.

Trainer-Class: A one-to-many relationship where one trainer can lead multiple classes. A single trainer can lead multiple classes, but each class is led by only one trainer.

Trainer-Session: A one-to-many relationship where each session is conducted by a single trainer, but a trainer can lead multiple sessions for different classes.

Trainer-Facility: A one-to-many relationship where a trainer may use different facilities for conducting sessions or personal training, but each facility can host sessions by multiple trainers.

Trainer-Workout Plan: A one-to-many relationship where each workout plan is designed by a single trainer, but a trainer can create multiple workout plans for different members.

Trainer-Trainer Certification: A one-to-many relationship where each trainer can hold multiple certifications, but each certification is linked to one trainer.

Trainer-Payment: A one-to-many relationship tracks the financial contribution of each trainer to the club's revenue. Payments from members are linked to the trainers leading the classes or sessions.

Trainer-Schedule: A one-to-one relationship where each trainer has a schedule that defines their availability and class assignments. Each schedule can include multiple sessions or classes, but each trainer has one schedule at a time.

Trainer-Referral: A one-to-many relationship where a trainer may be referred by other trainers or members. This can track how new trainers join the club or get recommended for specific classes or sessions.

Equipment-Class: A many-to-many relationship where multiple classes may use the same equipment, and each piece of equipment can be used by different classes.

Class-Equipment: A many-to-many relationship tracks the equipment required for each class. Each class uses different equipment, and the same equipment can be used in multiple classes.

Class-Payment: A many-to-one relationship tracks specific payments related to individual class enrolments. Members might pay separately for certain classes, even if they have a membership.

Class-Attendance: A one-to-many relationship where each class has multiple attendance records. Each record tracks which members attended that particular class at a specific time.

Promotion-Subscription: A one-to-many relationship where multiple subscriptions can be linked to a promotion. Each promotion applies to different membership plans or services.

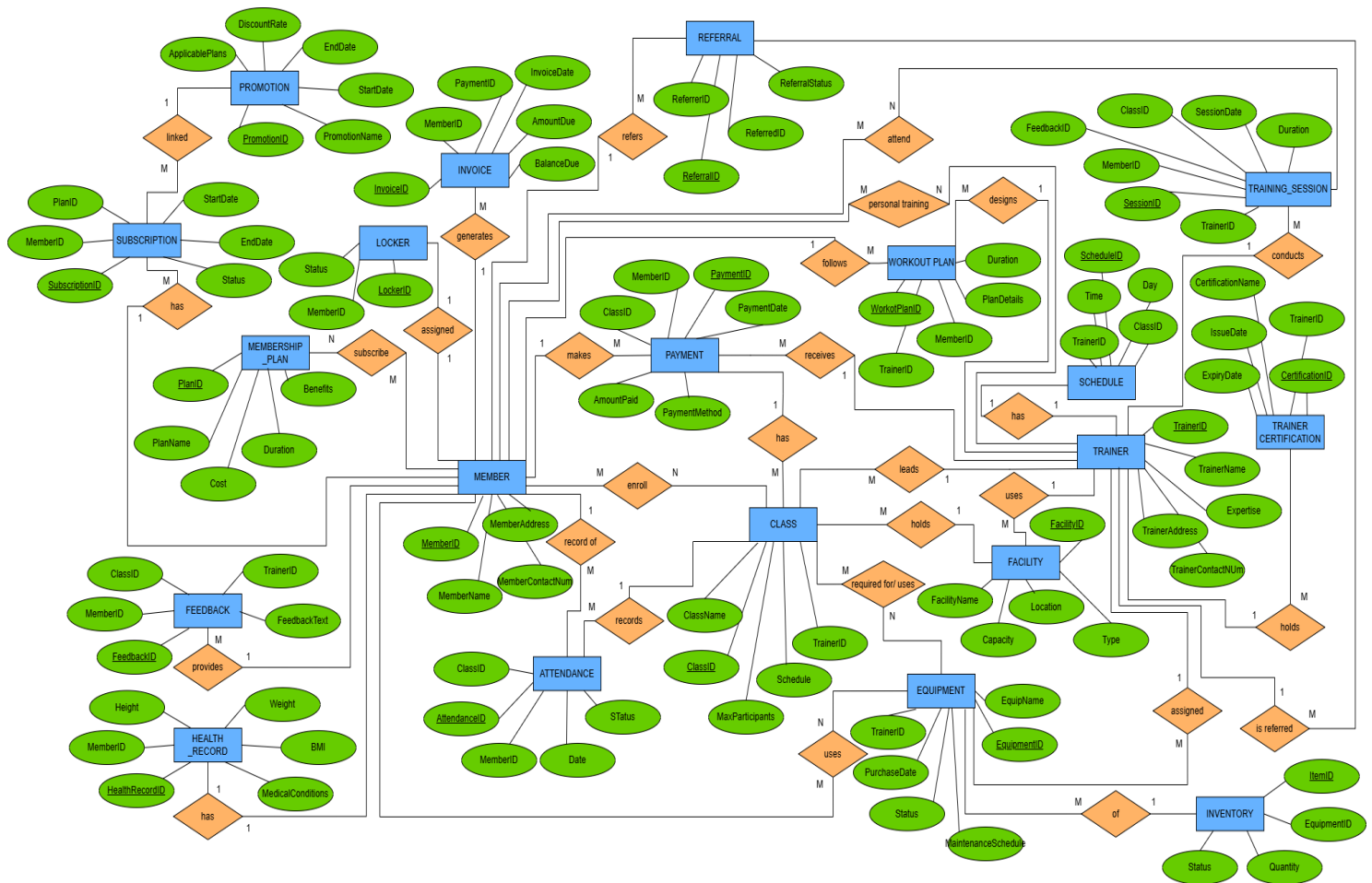
Inventory-Equipment: A one-to-many relationship where inventory tracks the availability and condition of various equipment items. Each piece of equipment is linked to the inventory.

Facility-Class: A one-to-many relationship where each class takes place in a specific facility. Each facility can host multiple classes, but each class occurs in only one facility at a time.

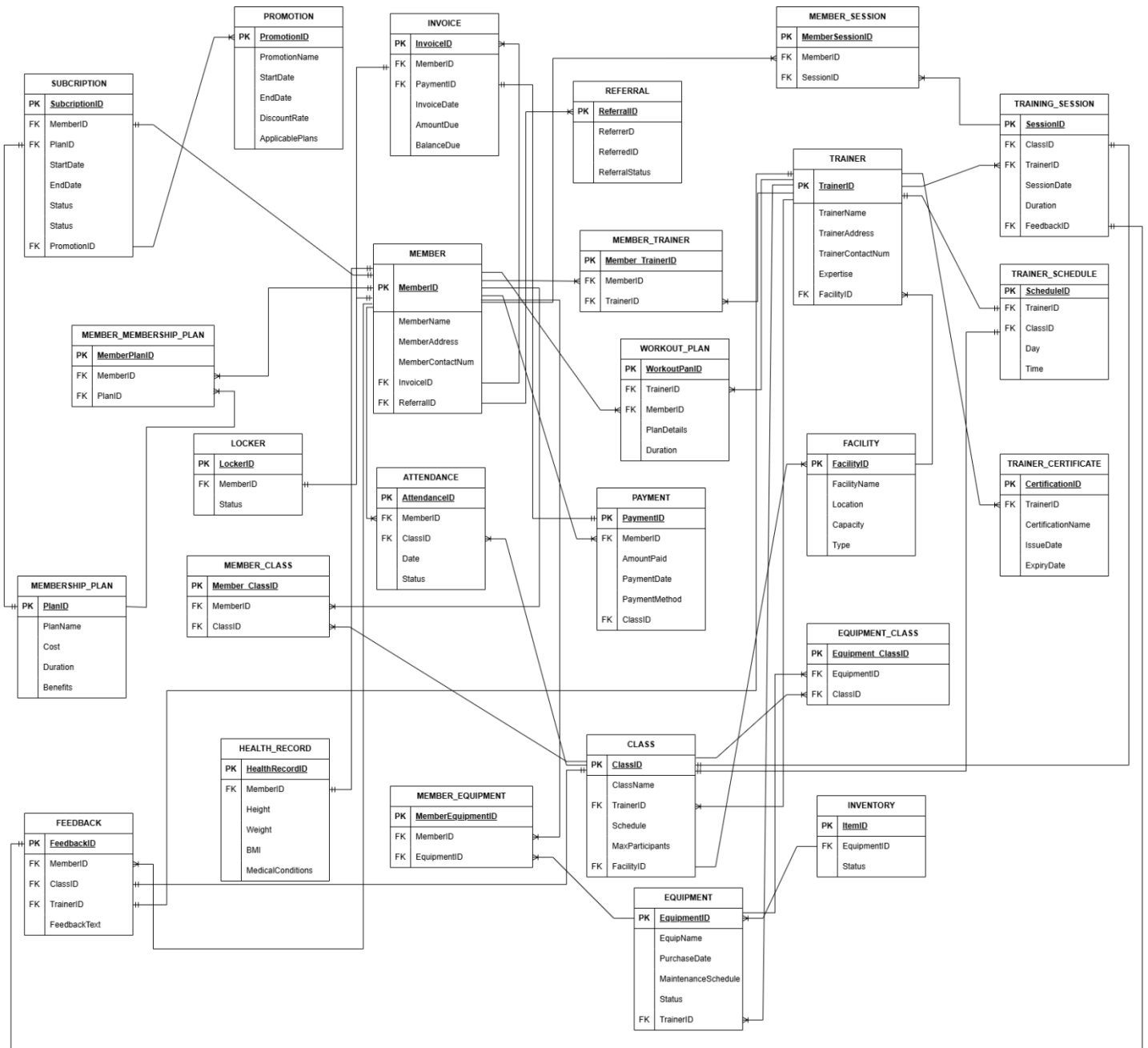
2.0 Database Design

2.1 Entity-Relationship Diagram (ERD)

CONCEPTUAL ERD



LOGICAL ERD



2.2 Mapping to Relational Schema

MEMBER(MemberID, MemberName, MemberAddress, MemberContactNum)

TRAINER(TrainerID, TrainerName, TrainerAddress, TrainerContactNum, Expertise)

CLASS(ClassID, ClassName, TrainerID, Schedule, MaxParticipants)

PAYMENT(PaymentID, MemberID, AmountPaid, PaymentDate, PaymentMethod, ClassID)

EQUIPMENT(EquipmentID, EquipName, PurchaseDate, MaintenanceSchedule, Status, TrainerID)

MEMBERSHIP_PLAN(PlanID, PlanName, Cost, Duration, Benefits)

FACILITY(FacilityID, FacilityName, Location, Capacity, Type)

ATTENDANCE(AttendanceID, MemberID, ClassID, Date, Status)

WORKOUT_PLAN(WorkoutPlanID, TrainerID, MemberID, PlanDetails, Duration)

TRAINER_CERTIFICATION(CertificationID, TrainerID, CertificationName, IssueDate, ExpiryDate)

HEALTH_RECORD(HealthRecordID, MemberID, Height, Weight, BMI, MedicalConditions)

FEEDBACK(FeedbackID, MemberID, ClassID, TrainerID, FeedbackText)

SUBSCRIPTION(SubscriptionID, MemberID, PlanID, StartDate, EndDate, Status)

PROMOTION(PromotionID, PromotionName, StartDate, EndDate, DiscountRate, ApplicablePlans)

SESSION(SessionID, TrainerID, MemberID, ClassID, SessionDate, Duration, FeedbackID)

INVOICE(InvoiceID, MemberID, PaymentID, InvoiceDate, AmountDue, BalanceDue)

LOCKER(LockerID, MemberID, Status)

REFERRAL(ReferralID, ReferrerID, ReferredID, ReferralStatus)

INVENTORY(ItemID, EquipmentID, Quantity, Status)

TRAINER_SCHEDULE(ScheduleID, TrainerID, ClassID, Day, Time)

2.3 Data Dictionary

MEMBER Table

Attribute Name	Data Type	Description	Constraints
MemberID	INT	Unique identifier for each member	PRIMARY KEY, AUTO_INCREMENT
MemberName	VARCHAR2(100)	Member's full name	NOT NULL
MemberAddress	VARCHAR2(200)	Member's address	
MemberContactNum	VARCHAR2(15)	Member's phone number	

TRAINER Table

Attribute Name	Data Type	Description	Constraints
TrainerID	INT	Unique identifier for each trainer	PRIMARY KEY, AUTO_INCREMENT
TrainerName	VARCHAR2(100)	Trainer's full name	NOT NULL
TrainerAddress	VARCHAR2(200)	Trainer's address	
TrainerContactNum	VARCHAR2(15)	Trainer's phone number	
Expertise	VARCHAR2(50)	Areas of expertise for the trainer	

CLASS Table

Attribute Name	Data Type	Description	Constraints
ClassID	INT	Unique identifier for each class	PRIMARY KEY, AUTO_INCREMENT
ClassName	VARCHAR2(100)	Name of the class	NOT NULL
TrainerId	INT	Identifier for the trainer leading the class	FOREIGN KEY (references TRAINER)
Schedule	DATE	Class schedule date and time	
MaxParticipants	INT	Maximum number of participants allowed	

PAYMENT Table

Attribute Name	Data Type	Description	Constraints
PaymentID	INT	Unique identifier for each payment	PRIMARY KEY, AUTO_INCREMENT
MemberID	INT	Identifier for the member making the payment	FOREIGN KEY (references MEMBER)
AmountPaid	DECIMAL(10,2)	Amount paid	
PaymentDate	DATE	Date and time of the payment	
PaymentMethod	VARCHAR2(20)	Method of payment	
ClassID	INT	Identifier for the class related to the payment	FOREIGN KEY (references CLASS)

EQUIPMENT Table

Attribute Name	Data Type	Description	Constraints
EquipmentID	INT	Unique identifier for each payment	PRIMARY KEY, AUTO_INCREMENT
EquipName	VARCHAR2(100)	Name of the equipment	NOT NULL
PurchaseDate	DATE	Date of purchase	
MaintenanceSchedule	VARCHAR2(50)	Scheduled maintenance date	
Status	VARCHAR2(20)	Current status of the equipment	
TrainerID	INT	Identifier for the trainer responsible for the equipment	FOREIGN KEY (references TRAINER)

MEMBERSHIP_PLAN Table

Attribute Name	Data Type	Description	Constraints
PlanID	INT	Unique identifier for each membership plan	PRIMARY KEY, AUTO_INCREMENT
PlanName	VARCHAR2(50)	Name of the membership plan	NOT NULL
Cost	DECIMAL(10,2)	Cost of the membership plan	
Duration	INT	Duration of the membership in months	
Benefits	VARCHAR2(200)	Benefits included in the membership plan	

FACILITY Table

Attribute Name	Data Type	Description	Constraints
FacilityID	INT	Unique identifier for each facility	PRIMARY KEY, AUTO_INCREMENT
FacilityName	VARCHAR2(100)	Name of the facility	NOT NULL
Location	VARCHAR2(100)	Location of the facility	
Capacity	INT	Maximum capacity of the facility	
Type	VARCHAR2(50)	Type of the facility	

ATTENDANCE Table

Attribute Name	Data Type	Description	Constraints
AttendanceID	INT	Unique identifier for each attendance record	PRIMARY KEY, AUTO_INCREMENT
MemberID	INT	Identifier for the member attending the class	FOREIGN KEY (references MEMBER)
ClassID	INT	Identifier for the class attended	FOREIGN KEY (references CLASS)
Date	DATE	Date of attendance	
Status	VARCHAR2(20)	Attendance status	

WORKOUT_PLAN Table

Attribute Name	Data Type	Description	Constraints
WorkoutPlanID	INT	Unique identifier for each workout plan	PRIMARY KEY, AUTO_INCREMENT
TrainerID	INT	Identifier for the trainer creating the plan	FOREIGN KEY (references TRAINER)
MemberID	INT	Identifier for the member receiving the plan	FOREIGN KEY (references MEMBER)
PlanDetails	VARCHAR2(200)	Details of the workout plan	
Duration	INT	Duration of the workout plan in weeks	

TRAINER_CERTIFICATION Table

Attribute Name	Data Type	Description	Constraints
CertificationID	INT	Unique identifier for each certification	PRIMARY KEY, AUTO_INCREMENT
TrainerID	INT	Identifier for the trainer	FOREIGN KEY (references TRAINER)
CertificationName	VARCHAR2(100)	Name of the certification	
IssueDate	DATE	Date the certification was issued	
ExpiryDate	DATE	Expiry date of the certification	

HEALTH_RECORD Table

Attribute Name	Data Type	Description	Constraints
HealthRecordID	INT	Unique identifier for each health record	PRIMARY KEY, AUTO_INCREMENT
MemberID	INT	Identifier for the member	FOREIGN KEY (references MEMBER)
Height	INT	Height of the member (in meters)	
Weight	INT	Weight of the member (in kg)	
BMI	DECIMAL(5,2)	Body Mass Index of the member	
MedicalConditions	VARCHAR2(200)	Medical conditions of the member	

FEEDBACK Table

Attribute Name	Data Type	Description	Constraints
FeedbackID	INT	Unique identifier for each feedback	PRIMARY KEY, AUTO_INCREMENT
MemberID	INT	Identifier for the member giving feedback	FOREIGN KEY (references MEMBER)
ClassID	INT	Identifier for the class being reviewed	FOREIGN KEY (references CLASS)
TrainerID	INT	Identifier for the trainer being reviewed	FOREIGN KEY (references TRAINER)
FeedbackText	VARCHAR2(500)	Feedback comments from the member	

SUBSCRIPTION Table

Attribute Name	Data Type	Description	Constraints
SubscriptionID	INT	Unique identifier for each subscription	PRIMARY KEY, AUTO_INCREMENT
MemberID	INT	Identifier for the member subscribing	FOREIGN KEY (references MEMBER)
PlanID	INT	Identifier for the subscribed plan	FOREIGN KEY (references MEMBERSHIP_PLAN)
StartDate	DATE	Start date of the subscription	
EndDate	DATE	End date of the subscription	
Status	VARCHAR2(20)	Status of the subscription	

PROMOTION Table

Attribute Name	Data Type	Description	Constraints
PromotionID	INT	Unique identifier for each promotion	PRIMARY KEY, AUTO_INCREMENT
PromotionName	VARCHAR2(100)	Name of the promotion	
StartDate	DATE	Start date of the promotion	
EndDate	DATE	End date of the promotion	
DiscountRate	DECIMAL(5,2)	Discount rate applied during the promotion	
ApplicablePlans	VARCHAR2(100)	Membership plans eligible for the promotion	

TRAINING_SESSION Table

Attribute Name	Data Type	Description	Constraints
SessionID	INT	Unique identifier for each session	PRIMARY KEY, AUTO_INCREMENT
TrainerID	INT	Identifier for the trainer leading the session	FOREIGN KEY (references TRAINER)
MemberID	INT	Identifier for the member attending the session	FOREIGN KEY (references MEMBER)
ClassID	INT	Identifier for the class associated with the session	FOREIGN KEY (references CLASS)
SessionDate	DATE	Date and time of the session	
Duration	INT	Duration of the session in minute	
FeedbackID	INT	Identifier for feedback related to the session	FOREIGN KEY (references FEEDBACK)

INVOICE Table

Attribute Name	Data Type	Description	Constraints
InvoiceID	VARCHAR2(100)	Unique identifier for each invoice	PRIMARY KEY, AUTO_INCREMENT
MemberID	NUMBER(10)	Identifier for the member associated with the invoice	FOREIGN KEY (references MEMBER)
PaymentID	NUMBER(10)	Identifier for the payment associated with the invoice	FOREIGN KEY (references PAYMENT)
InvoiceDate	DATE	Date of invoice issuance	
AmountDue	DECIMAL(10,2)	Total amount due	
BalanceDue	DECIMAL(10,2)	Balance due if partial payment made	

LOCKER Table

Attribute Name	Data Type	Description	Constraints
LockerID	VARCHAR2(10)	Unique identifier for each locker	PRIMARY KEY
MemberID	INT	Identifier for the member using the locker	FOREIGN KEY (references MEMBER)
Status	VARCHAR2(20)	Status of the locker	

REFFERAL Table

Attribute Name	Data Type	Description	Constraints
ReferralID	VARCHAR2(10)	Unique identifier for each referral	PRIMARY KEY, AUTO_INCREMENT
ReferrerID	NUMBER(10)	Identifier for the member who referred another member	FOREIGN KEY (references MEMBER)
ReferredID	NUMBER(10)	Identifier for the referred member	FOREIGN KEY (references MEMBER)
ReferralStatus	VARCHAR2(20)	Status of the referral	

INVENTORY Table

Attribute Name	Data Type	Description	Constraints
ItemID	VARCHAR2(10)	Unique identifier for each inventory item	PRIMARY KEY, AUTO_INCREMENT
EquipmenID	NUMBER(10)	Identifier for the related equipment	FOREIGN KEY (references EQUIPMENT)
Quantity	NUMBER(5)	Quantity of the item in stock	
Status	VARCHAR2(20)	Status of the inventory item	

TRAINER_SCHEDULE Table

Attribute Name	Data Type	Description	Constraints
ScheduleID	INT	Unique identifier for each scedule	PRIMARY KEY, AUTO_INCREMENT
TrainerID	INT	Identifier for the trainer	FOREIGN KEY (references TRAINER)
ClassID	INT	Identifier for the class in the schedule	FOREIGN KEY (references CLASS)
Day	VARACHAR2(20)	Day of the week for the scheduled class	
Time	VARACHAR2(20)	Time of the scheduled class	

3.0 Database Implementation

3.1 SQL Code for Database Creation

-- Create MEMBER table

```
CREATE TABLE MEMBER (  
    MemberID INT PRIMARY KEY,  
    MemberName VARCHAR2(100) NOT NULL,  
    MemberAddress VARCHAR2(200),  
    MemberContactNum VARCHAR2(15)  
);
```

-- Create TRAINER table

```
CREATE TABLE TRAINER (  
    TrainerID INT PRIMARY KEY,  
    TrainerName VARCHAR2(100) NOT NULL,  
    TrainerAddress VARCHAR2(200),  
    TrainerContactNum VARCHAR2(15),  
    Expertise VARCHAR2(50)  
);
```

-- Create CLASS table

```
CREATE TABLE CLASS (  
    ClassID INT PRIMARY KEY,  
    ClassName VARCHAR2(100) NOT NULL,  
    TrainerID INT,  
    Schedule DATE,  
    MaxParticipants INT,  
    FOREIGN KEY (TrainerID) REFERENCES TRAINER(TrainerID) ON DELETE CASCADE  
);
```

-- Create PAYMENT table

```
CREATE TABLE PAYMENT (  
    PaymentID INT PRIMARY KEY,  
    MemberID INT,  
    AmountPaid DECIMAL(10, 2),  
    PaymentDate DATE,  
    PaymentMethod VARCHAR2(20),  
    ClassID INT,  
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE,  
    FOREIGN KEY (ClassID) REFERENCES CLASS(ClassID) ON DELETE CASCADE  
);
```

-- Create EQUIPMENT table

```
CREATE TABLE EQUIPMENT (  
    EquipmentID INT PRIMARY KEY,  
    EquipName VARCHAR2(100) NOT NULL,  
    PurchaseDate DATE,  
    MaintenanceSchedule VARCHAR2(50),  
    Status VARCHAR2(20),  
    TrainerID INT,  
    FOREIGN KEY (TrainerID) REFERENCES TRAINER(TrainerID) ON DELETE CASCADE  
);
```

-- Create MEMBERSHIP_PLAN table

```
CREATE TABLE MEMBERSHIP_PLAN (  
    PlanID INT PRIMARY KEY,  
    PlanName VARCHAR2(50) NOT NULL,  
    Cost DECIMAL(10, 2),  
    Duration INT,  
    Benefits VARCHAR2(200)  
);
```

-- Create FACILITY table

```
CREATE TABLE FACILITY (  
    FacilityID INT PRIMARY KEY,  
    FacilityName VARCHAR2(100) NOT NULL,  
    Location VARCHAR2(100),  
    Capacity INT,  
    Type VARCHAR2(50)  
);
```

-- Create ATTENDANCE table

```
CREATE TABLE ATTENDANCE (  
    AttendanceID INT PRIMARY KEY,  
    MemberID INT,  
    ClassID INT,  
    AttendanceDate DATE,  
    AttendanceStatus VARCHAR2(20),  
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE,  
    FOREIGN KEY (ClassID) REFERENCES CLASS(ClassID) ON DELETE CASCADE  
);
```

-- Create WORKOUT_PLAN table

```
CREATE TABLE WORKOUT_PLAN (  
    WorkoutPlanID INT PRIMARY KEY,  
    TrainerID INT,  
    MemberID INT,  
    PlanDetails VARCHAR2(200),  
    Duration INT,  
    FOREIGN KEY (TrainerID) REFERENCES TRAINER(TrainerID) ON DELETE CASCADE,  
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE  
);
```

-- Create TRAINER_CERTIFICATION table

```
CREATE TABLE TRAINER_CERTIFICATION (  
    CertificationID INT PRIMARY KEY,  
    TrainerID INT,  
    CertificationName VARCHAR2(100),  
    IssueDate DATE,  
    ExpiryDate DATE,  
    FOREIGN KEY (TrainerID) REFERENCES TRAINER(TrainerID) ON DELETE CASCADE  
);
```

-- Create HEALTH_RECORD table

```
CREATE TABLE HEALTH_RECORD (  
    HealthRecordID INT PRIMARY KEY,  
    MemberID INT,  
    Height INT,  
    Weight INT,  
    BMI DECIMAL(5, 2),  
    MedicalConditions VARCHAR2(200),  
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE  
);
```

-- Create FEEDBACK table

```
CREATE TABLE FEEDBACK (  
    FeedbackID INT PRIMARY KEY,  
    MemberID INT,  
    ClassID INT,  
    TrainerID INT,  
    FeedbackText VARCHAR2(500),  
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE,  
    FOREIGN KEY (ClassID) REFERENCES CLASS(ClassID) ON DELETE CASCADE,  
    FOREIGN KEY (TrainerID) REFERENCES TRAINER(TrainerID) ON DELETE CASCADE  
);
```

-- Create SUBSCRIPTION table

```
CREATE TABLE SUBSCRIPTION (  
    SubscriptionID INT PRIMARY KEY,  
    MemberID INT,  
    PlanID INT,  
    StartDate DATE,  
    EndDate DATE,  
    Status VARCHAR2(20),  
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE,  
    FOREIGN KEY (PlanID) REFERENCES MEMBERSHIP_PLAN(PlanID) ON DELETE CASCADE  
);
```

-- Create PROMOTION table

```
CREATE TABLE PROMOTION (  
    PromotionID INT PRIMARY KEY,  
    PromotionName VARCHAR2(100),  
    StartDate DATE,  
    EndDate DATE,  
    DiscountRate DECIMAL(5, 2),  
    ApplicablePlans VARCHAR2(100)  
);
```

-- Create TRAINING_SESSION table

```
CREATE TABLE TRAINING_SESSION (  
    SessionID INT PRIMARY KEY,  
    TrainerID INT,  
    MemberID INT,  
    ClassID INT,  
    SessionDate DATE,  
    Duration INT,  
    FeedbackID INT,  
    FOREIGN KEY (TrainerID) REFERENCES TRAINER(TrainerID) ON DELETE CASCADE,  
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE,  
    FOREIGN KEY (ClassID) REFERENCES CLASS(ClassID) ON DELETE CASCADE,  
    FOREIGN KEY (FeedbackID) REFERENCES FEEDBACK(FeedbackID) ON DELETE CASCADE  
);
```

-- Create INVOICE table

```
CREATE TABLE INVOICE (  
    InvoiceID VARCHAR2(10) PRIMARY KEY,  
    MemberID NUMBER(10),  
    PaymentID NUMBER(10),  
    InvoiceDate DATE,  
    AmountDue DECIMAL(10, 2),  
    BalanceDue DECIMAL(10, 2),  
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE,  
    FOREIGN KEY (PaymentID) REFERENCES PAYMENT(PaymentID) ON DELETE CASCADE  
);
```

-- Create LOCKER table

```
CREATE TABLE LOCKER (  
    LockerID VARCHAR2(10) PRIMARY KEY,  
    MemberID INT,  
    Status VARCHAR2(20),  
    FOREIGN KEY (MemberID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE  
);
```

-- Create REFFERRAL table

```
CREATE TABLE REFERRAL (  
  ReferralID VARCHAR2(10) PRIMARY KEY,  
  ReferrerID NUMBER(10),  
  ReferredID NUMBER(10),  
  ReferralStatus VARCHAR2(20),  
  FOREIGN KEY (ReferrerID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE,  
  FOREIGN KEY (ReferredID) REFERENCES MEMBER(MemberID) ON DELETE CASCADE  
);
```

-- Create INVENTORY table

```
CREATE TABLE INVENTORY (  
  ItemID VARCHAR2(10) PRIMARY KEY,  
  EquipmentID NUMBER(10),  
  Quantity NUMBER(5),  
  Status VARCHAR2(20),  
  FOREIGN KEY (EquipmentID) REFERENCES EQUIPMENT(EquipmentID) ON DELETE CASCADE  
);
```

-- Create TRAINER_SCHEDULE table

```
CREATE TABLE TRAINER_SCHEDULE (  
  ScheduleID INT PRIMARY KEY,  
  TrainerID INT,  
  ClassID INT,  
  Day VARCHAR2(20),  
  Time VARCHAR2(20),  
  FOREIGN KEY (TrainerID) REFERENCES TRAINER(TrainerID) ON DELETE CASCADE,  
  FOREIGN KEY (ClassID) REFERENCES CLASS(ClassID) ON DELETE CASCADE  
);
```

3.2 Database Population

-- Insert into MEMBER table

```
INSERT INTO MEMBER (MemberID, MemberName, MemberAddress, MemberContactNum)
VALUES (1001, 'Ali Ahmad', '123 Green Street', '012-3456789');
INSERT INTO MEMBER (MemberID, MemberName, MemberAddress, MemberContactNum)
VALUES (1002, 'Farah Binti Ismail', '456 Blue Avenue', '019-8765432');
INSERT INTO MEMBER (MemberID, MemberName, MemberAddress, MemberContactNum)
VALUES (1003, 'John Doe', '789 Red Road', '013-5674321');
INSERT INTO MEMBER (MemberID, MemberName, MemberAddress, MemberContactNum)
VALUES (1004, 'Zainab Abdullah', '321 Yellow Lane', '017-2349876') ;
INSERT INTO MEMBER (MemberID, MemberName, MemberAddress, MemberContactNum)
VALUES (1005, 'Karim Bin Hassan', '654 Orange Crescent', '016-9834567') ;
```

-- Insert into TRAINER table

```
INSERT INTO TRAINER (TrainerID, TrainerName, TrainerAddress, TrainerContactNum,
Expertise)
VALUES (3001, 'Haziq Hassan', '101 Maple Drive', '018-7654321', 'Yoga') ;
INSERT INTO TRAINER (TrainerID, TrainerName, TrainerAddress, TrainerContactNum,
Expertise)
VALUES (3002, 'Sarah Lim', '202 Oak Avenue', '019-5432187', 'Cardio') ;
INSERT INTO TRAINER (TrainerID, TrainerName, TrainerAddress, TrainerContactNum,
Expertise)
VALUES (3003, 'Ahmad Syukri', '303 Pine Street', '017-8901234', 'Pilates');
INSERT INTO TRAINER (TrainerID, TrainerName, TrainerAddress, TrainerContactNum,
Expertise)
VALUES (3004, 'Emily Wong', '404 Cedar Road', '012-3450912', 'Strength') ;
INSERT INTO TRAINER (TrainerID, TrainerName, TrainerAddress, TrainerContactNum,
Expertise)
VALUES (3005, 'Aminah Ali', '505 Birch Crescent', '016-2348765', 'Zumba');
```

-- Insert into CLASS table

```
INSERT INTO CLASS (ClassID, ClassName, TrainerID, Schedule, MaxParticipants)
VALUES (4001, 'Zumba', 3005, TO_DATE('10/01/2024 10:00 AM', 'DD/MM/YYYY HH:MI AM'),
20) ;
INSERT INTO CLASS (ClassID, ClassName, TrainerID, Schedule, MaxParticipants)
VALUES (4002, 'Pilates', 3003, TO_DATE('12/01/2024 12:00 PM', 'DD/MM/YYYY HH:MI
AM'), 15);
INSERT INTO CLASS (ClassID, ClassName, TrainerID, Schedule, MaxParticipants)
VALUES (4003, 'Yoga', 3001, TO_DATE('15/01/2024 09:00 AM', 'DD/MM/YYYY HH:MI AM'),
25) ;
INSERT INTO CLASS (ClassID, ClassName, TrainerID, Schedule, MaxParticipants)
VALUES (4004, 'Cardio', 3002, TO_DATE('16/01/2024 11:00 AM', 'DD/MM/YYYY HH:MI
AM'), 30) ;
INSERT INTO CLASS (ClassID, ClassName, TrainerID, Schedule, MaxParticipants)
VALUES (4005, 'Strength', 3004, TO_DATE('17/01/2024 02:00 PM', 'DD/MM/YYYY HH:MI
AM'), 15);
```

-- Insert into PAYMENT table

```
INSERT INTO PAYMENT (PaymentID, MemberID, AmountPaid, PaymentDate, PaymentMethod,
ClassID)
VALUES (5001, 1001, 1200, TO_DATE('05/01/2024', 'DD/MM/YYYY'), 'Credit Card',
4001);
INSERT INTO PAYMENT (PaymentID, MemberID, AmountPaid, PaymentDate, PaymentMethod,
ClassID)
VALUES (5002, 1002, 100, TO_DATE('06/01/2024', 'DD/MM/YYYY'), 'Cash', 4002);
INSERT INTO PAYMENT (PaymentID, MemberID, AmountPaid, PaymentDate, PaymentMethod,
ClassID)
VALUES (5003, 1003, 1500, TO_DATE('07/01/2024', 'DD/MM/YYYY'), 'Debit Card', 4003);
INSERT INTO PAYMENT (PaymentID, MemberID, AmountPaid, PaymentDate, PaymentMethod,
ClassID)
VALUES (5004, 1004, 150, TO_DATE('08/01/2024', 'DD/MM/YYYY'), 'Cash', 4004);
INSERT INTO PAYMENT (PaymentID, MemberID, AmountPaid, PaymentDate, PaymentMethod,
ClassID)
VALUES (5005, 1005, 600, TO_DATE('09/01/2024', 'DD/MM/YYYY'), 'Credit Card', 4005);
```

-- Insert into EQUIPMENT table

```
INSERT INTO EQUIPMENT (EquipmentID, EquipName, PurchaseDate, MaintenanceSchedule,
Status, TrainerID)
VALUES (2001, 'Treadmill', TO_DATE('01/06/2023', 'DD/MM/YYYY'), 'Monthly',
'Available', 3001);
INSERT INTO EQUIPMENT (EquipmentID, EquipName, PurchaseDate, MaintenanceSchedule,
Status, TrainerID)
VALUES (2002, 'Dumbbells', TO_DATE('15/07/2023', 'DD/MM/YYYY'), 'Quarterly', 'In
Use', 3002) ;
INSERT INTO EQUIPMENT (EquipmentID, EquipName, PurchaseDate, MaintenanceSchedule,
Status, TrainerID)
VALUES (2003, 'Rowing Machine', TO_DATE('20/05/2023', 'DD/MM/YYYY'), 'Monthly',
'Available', 3003);
INSERT INTO EQUIPMENT (EquipmentID, EquipName, PurchaseDate, MaintenanceSchedule,
Status, TrainerID)
VALUES (2004, 'Exercise Bike', TO_DATE('30/06/2023', 'DD/MM/YYYY'), 'Monthly',
'Available', 3004);
INSERT INTO EQUIPMENT (EquipmentID, EquipName, PurchaseDate, MaintenanceSchedule,
Status, TrainerID)
VALUES (2005, 'Kettlebells', TO_DATE('25/04/2023', 'DD/MM/YYYY'), 'Quarterly',
'Available', 3005);
```

-- Insert into MEMBERSHIP_PLAN table

```
INSERT INTO MEMBERSHIP_PLAN (PlanID, PlanName, Cost, Duration, Benefits)
VALUES (6001, 'Basic', 500, 1, 'Access to gym, 1 class per week');
INSERT INTO MEMBERSHIP_PLAN (PlanID, PlanName, Cost, Duration, Benefits)
VALUES (6002, 'Premium', 1200, 12, 'Unlimited classes, personal training');
INSERT INTO MEMBERSHIP_PLAN (PlanID, PlanName, Cost, Duration, Benefits)
VALUES (6003, 'Family', 3000, 12, 'Access for 4 members, free equipment');
INSERT INTO MEMBERSHIP_PLAN (PlanID, PlanName, Cost, Duration, Benefits)
VALUES (6004, 'Student', 300, 1, 'Discounted rates, 2 classes per week');
INSERT INTO MEMBERSHIP_PLAN (PlanID, PlanName, Cost, Duration, Benefits)
VALUES (6005, 'VIP', 5000, 12, 'All-access pass, free workshops');
```

-- Insert into FACILITY table

```
INSERT INTO FACILITY (FacilityID, FacilityName, Location, Capacity, Type)
VALUES (7001, 'Gym Floor', 'Ground Floor', 50, 'Gym');
INSERT INTO FACILITY (FacilityID, FacilityName, Location, Capacity, Type)
VALUES (7002, 'Yoga Studio', 'Second Floor', 25, 'Studio');
INSERT INTO FACILITY (FacilityID, FacilityName, Location, Capacity, Type)
VALUES (7003, 'Pool', 'Outdoor Area', 20, 'Pool') ;
INSERT INTO FACILITY (FacilityID, FacilityName, Location, Capacity, Type)
VALUES (7004, 'Cardio Room', 'Front Floor', 30, 'Gym') ;
INSERT INTO FACILITY (FacilityID, FacilityName, Location, Capacity, Type)
VALUES (7005, 'Spin Room', 'Third Floor', 15, 'Studio');
```

-- Insert into ATTENDANCE table

```
INSERT INTO ATTENDANCE (AttendanceID, MemberID, ClassID, AttendanceDate,
AttendanceStatus)
VALUES (8001, 1001, 4001, TO_DATE('05/01/2024', 'DD/MM/YYYY'), 'Present');
INSERT INTO ATTENDANCE (AttendanceID, MemberID, ClassID, AttendanceDate,
AttendanceStatus)
VALUES (8002, 1002, 4002, TO_DATE('06/01/2024', 'DD/MM/YYYY'), 'Absent') ;
INSERT INTO ATTENDANCE (AttendanceID, MemberID, ClassID, AttendanceDate,
AttendanceStatus)
VALUES (8003, 1003, 4003, TO_DATE('07/01/2024', 'DD/MM/YYYY'), 'Present') ;
INSERT INTO ATTENDANCE (AttendanceID, MemberID, ClassID, AttendanceDate,
AttendanceStatus)
VALUES (8004, 1004, 4004, TO_DATE('08/01/2024', 'DD/MM/YYYY'), 'Present');
INSERT INTO ATTENDANCE (AttendanceID, MemberID, ClassID, AttendanceDate,
AttendanceStatus)
VALUES (8005, 1005, 4005, TO_DATE('09/01/2024', 'DD/MM/YYYY'), 'Absent');
```

-- Insert into WORKOUT_PLAN table

```
INSERT INTO WORKOUT_PLAN (WorkoutPlanID, TrainerID, MemberID, PlanDetails,
Duration)
VALUES (9001, 3001, 1001, 'Yoga & Cardio Routine', 4);
INSERT INTO WORKOUT_PLAN (WorkoutPlanID, TrainerID, MemberID, PlanDetails,
Duration)
VALUES (9002, 3002, 1002, 'Strength & Endurance', 6) ;
INSERT INTO WORKOUT_PLAN (WorkoutPlanID, TrainerID, MemberID, PlanDetails,
Duration)
VALUES (9003, 3003, 1003, 'Pilates & Core Training', 8);
INSERT INTO WORKOUT_PLAN (WorkoutPlanID, TrainerID, MemberID, PlanDetails,
Duration)
VALUES (9004, 3004, 1004, 'Weight Loss & Toning', 12);
INSERT INTO WORKOUT_PLAN (WorkoutPlanID, TrainerID, MemberID, PlanDetails,
Duration)
VALUES (9005, 3005, 1005, 'Zumba & Dance Fitness', 10);
```

-- Insert into TRAINER_CERTIFICATION table

```
INSERT INTO TRAINER_CERTIFICATION (CertificationID, TrainerID, CertificationName,
IssueDate, ExpiryDate)
VALUES (10001, 3001, 'Yoga Instructor', TO_DATE('01/01/2023', 'DD/MM/YYYY'),
TO_DATE('01/01/2026', 'DD/MM/YYYY')) ;
INSERT INTO TRAINER_CERTIFICATION (CertificationID, TrainerID, CertificationName,
IssueDate, ExpiryDate)
VALUES (10002, 3002, 'Cardio Specialist', TO_DATE('01/02/2023', 'DD/MM/YYYY'),
TO_DATE('01/02/2026', 'DD/MM/YYYY')) ;
INSERT INTO TRAINER_CERTIFICATION (CertificationID, TrainerID, CertificationName,
IssueDate, ExpiryDate)
VALUES (10003, 3003, 'Pilates Trainer', TO_DATE('05/03/2023', 'DD/MM/YYYY'),
TO_DATE('05/03/2026', 'DD/MM/YYYY')) ;
INSERT INTO TRAINER_CERTIFICATION (CertificationID, TrainerID, CertificationName,
IssueDate, ExpiryDate)
VALUES (10004, 3004, 'Strength Coach', TO_DATE('01/04/2023', 'DD/MM/YYYY'),
TO_DATE('01/04/2026', 'DD/MM/YYYY')) ;
INSERT INTO TRAINER_CERTIFICATION (CertificationID, TrainerID, CertificationName,
IssueDate, ExpiryDate)
VALUES (10005, 3005, 'Zumba Instructor', TO_DATE('01/05/2023', 'DD/MM/YYYY'),
TO_DATE('01/05/2026', 'DD/MM/YYYY')) ;
```

-- Insert into HEALTH_RECORD table

```
INSERT INTO HEALTH_RECORD (HealthRecordID, MemberID, Height, Weight, BMI,
MedicalConditions)
VALUES (11001, 1001, 170, 75, 25.9, 'None');
INSERT INTO HEALTH_RECORD (HealthRecordID, MemberID, Height, Weight, BMI,
MedicalConditions)
VALUES (11002, 1002, 160, 65, 25.4, 'Hypertension');
INSERT INTO HEALTH_RECORD (HealthRecordID, MemberID, Height, Weight, BMI,
MedicalConditions)
VALUES (11003, 1003, 180, 80, 24.7, 'Asthma') ;
INSERT INTO HEALTH_RECORD (HealthRecordID, MemberID, Height, Weight, BMI,
MedicalConditions)
VALUES (11004, 1004, 165, 55, 20.2, 'None');
INSERT INTO HEALTH_RECORD (HealthRecordID, MemberID, Height, Weight, BMI,
MedicalConditions)
VALUES (11005, 1005, 175, 90, 29.4, 'Diabetes');
```

-- Insert into FEEDBACK table

```
INSERT INTO FEEDBACK (FeedbackID, MemberID, ClassID, TrainerID, FeedbackText)
VALUES (12001, 1001, 4001, 3005, 'Great class, loved the energy!');
INSERT INTO FEEDBACK (FeedbackID, MemberID, ClassID, TrainerID, FeedbackText)
VALUES (12002, 1002, 4002, 3004, 'Would like more variations in routines.');
```

```
INSERT INTO FEEDBACK (FeedbackID, MemberID, ClassID, TrainerID, FeedbackText)
VALUES (12003, 1003, 4003, 3001, 'Very relaxing, felt refreshed!');
INSERT INTO FEEDBACK (FeedbackID, MemberID, ClassID, TrainerID, FeedbackText)
VALUES (12004, 1004, 4004, 3002, 'Good instructor but class was too crowded.');
```

```
INSERT INTO FEEDBACK (FeedbackID, MemberID, ClassID, TrainerID, FeedbackText)
VALUES (12005, 1005, 4005, 3003, 'Fun workout, would definitely come again!');
```

-- Insert into SUBSCRIPTION table

```
INSERT INTO SUBSCRIPTION (SubscriptionID, MemberID, PlanID, StartDate, EndDate, Status)
VALUES (13001, 1001, 6002, TO_DATE('01/01/2024', 'DD/MM/YYYY'),
TO_DATE('01/01/2025', 'DD/MM/YYYY'), 'Active') ;
INSERT INTO SUBSCRIPTION (SubscriptionID, MemberID, PlanID, StartDate, EndDate, Status)
VALUES (13002, 1002, 6004, TO_DATE('10/01/2024', 'DD/MM/YYYY'),
TO_DATE('10/02/2024', 'DD/MM/YYYY'), 'Expired');
INSERT INTO SUBSCRIPTION (SubscriptionID, MemberID, PlanID, StartDate, EndDate, Status)
VALUES (13003, 1003, 6001, TO_DATE('01/01/2024', 'DD/MM/YYYY'),
TO_DATE('01/02/2024', 'DD/MM/YYYY'), 'Expired') ;
INSERT INTO SUBSCRIPTION (SubscriptionID, MemberID, PlanID, StartDate, EndDate, Status)
VALUES (13004, 1004, 6005, TO_DATE('15/01/2024', 'DD/MM/YYYY'),
TO_DATE('15/01/2025', 'DD/MM/YYYY'), 'Active');
INSERT INTO SUBSCRIPTION (SubscriptionID, MemberID, PlanID, StartDate, EndDate, Status)
VALUES (13005, 1005, 6003, TO_DATE('15/01/2024', 'DD/MM/YYYY'),
TO_DATE('15/01/2025', 'DD/MM/YYYY'), 'Active');
```

-- Insert into PROMOTION table

```
INSERT INTO PROMOTION (PromotionID, PromotionName, StartDate, EndDate, DiscountRate, ApplicablePlans)
VALUES (14001, 'New Year Special', TO_DATE('01/01/2024', 'DD/MM/YYYY'),
TO_DATE('31/01/2024', 'DD/MM/YYYY'), 10, '6001, 6002, 6005') ;
INSERT INTO PROMOTION (PromotionID, PromotionName, StartDate, EndDate, DiscountRate, ApplicablePlans)
VALUES (14002, 'Student Discount', TO_DATE('01/02/2024', 'DD/MM/YYYY'),
TO_DATE('30/06/2023', 'DD/MM/YYYY'), 15, '6004') ;
INSERT INTO PROMOTION (PromotionID, PromotionName, StartDate, EndDate, DiscountRate, ApplicablePlans)
VALUES (14003, 'Family & Friends', TO_DATE('15/01/2024', 'DD/MM/YYYY'),
TO_DATE('15/02/2024', 'DD/MM/YYYY'), 20, '6003');
INSERT INTO PROMOTION (PromotionID, PromotionName, StartDate, EndDate, DiscountRate, ApplicablePlans)
VALUES (14004, 'Summer Fitness Deal', TO_DATE('01/06/2024', 'DD/MM/YYYY'),
TO_DATE('31/08/2023', 'DD/MM/YYYY'), 25, '6002, 6003, 6005');
INSERT INTO PROMOTION (PromotionID, PromotionName, StartDate, EndDate, DiscountRate, ApplicablePlans)
VALUES (14005, 'VIP Special Member', TO_DATE('01/04/2024', 'DD/MM/YYYY'),
TO_DATE('30/04/2024', 'DD/MM/YYYY'), 30, '6005');
```

-- Insert into TRAINING_SESSION table

```
INSERT INTO TRAINING_SESSION (SessionID, TrainerID, MemberID, ClassID, SessionDate,
Duration, FeedbackID)
VALUES (15001, 3001, 1001, 4001, TO_DATE('05/01/2024', 'DD/MM/YYYY'), 60, 12001);
INSERT INTO TRAINING_SESSION (SessionID, TrainerID, MemberID, ClassID, SessionDate,
Duration, FeedbackID)
VALUES (15002, 3002, 1002, 4002, TO_DATE('06/01/2024', 'DD/MM/YYYY'), 45, 12002);
INSERT INTO TRAINING_SESSION (SessionID, TrainerID, MemberID, ClassID, SessionDate,
Duration, FeedbackID)
VALUES (15003, 3003, 1003, 4003, TO_DATE('07/01/2024', 'DD/MM/YYYY'), 60, 12003);
INSERT INTO TRAINING_SESSION (SessionID, TrainerID, MemberID, ClassID, SessionDate,
Duration, FeedbackID)
VALUES (15004, 3004, 1004, 4004, TO_DATE('08/01/2024', 'DD/MM/YYYY'), 60, 12004);
INSERT INTO TRAINING_SESSION (SessionID, TrainerID, MemberID, ClassID, SessionDate,
Duration, FeedbackID)
VALUES (15005, 3005, 1005, 4005, TO_DATE('09/01/2024', 'DD/MM/YYYY'), 45, 12005);
```

-- Insert into INVOICE table

```
INSERT INTO INVOICE (InvoiceID, MemberID, PaymentID, InvoiceDate, AmountDue,
BalanceDue)
VALUES ('INV10001', 1001, 5001, TO_DATE('05/01/2024', 'DD/MM/YYYY'), 1200, 0);
INSERT INTO INVOICE (InvoiceID, MemberID, PaymentID, InvoiceDate, AmountDue,
BalanceDue)
VALUES ('INV10002', 1002, 5002, TO_DATE('06/01/2024', 'DD/MM/YYYY'), 100, 0);
INSERT INTO INVOICE (InvoiceID, MemberID, PaymentID, InvoiceDate, AmountDue,
BalanceDue)
VALUES ('INV10003', 1003, 5003, TO_DATE('07/01/2024', 'DD/MM/YYYY'), 1500, 300);
INSERT INTO INVOICE (InvoiceID, MemberID, PaymentID, InvoiceDate, AmountDue,
BalanceDue)
VALUES ('INV10004', 1004, 5004, TO_DATE('08/01/2024', 'DD/MM/YYYY'), 150, 0);
INSERT INTO INVOICE (InvoiceID, MemberID, PaymentID, InvoiceDate, AmountDue,
BalanceDue)
VALUES ('INV10005', 1005, 5005, TO_DATE('09/01/2024', 'DD/MM/YYYY'), 600, 0);
```

-- Insert into LOCKER table

```
INSERT INTO LOCKER (LockerID, MemberID, Status) VALUES ('L01', 1001, 'Assigned');
INSERT INTO LOCKER (LockerID, MemberID, Status) VALUES ('L02', 1002, 'Assigned');
INSERT INTO LOCKER (LockerID, MemberID, Status) VALUES ('L03', NULL, 'Available');
INSERT INTO LOCKER (LockerID, MemberID, Status) VALUES ('L04', 1004, 'Assigned');
INSERT INTO LOCKER (LockerID, MemberID, Status) VALUES ('L05', NULL, 'Available');
```

-- Insert into REFERRAL table

```
INSERT INTO REFERRAL (ReferralID, ReferrerID, ReferredID, ReferralStatus)
VALUES ('REF001', 1001, 1002, 'Accepted');
INSERT INTO REFERRAL (ReferralID, ReferrerID, ReferredID, ReferralStatus)
VALUES ('REF002', 1003, 1004, 'Pending');
INSERT INTO REFERRAL (ReferralID, ReferrerID, ReferredID, ReferralStatus)
VALUES ('REF003', 1004, 1005, 'Rejected');
INSERT INTO REFERRAL (ReferralID, ReferrerID, ReferredID, ReferralStatus)
VALUES ('REF004', 1001, 1003, 'Accepted');
INSERT INTO REFERRAL (ReferralID, ReferrerID, ReferredID, ReferralStatus)
VALUES ('REF005', 1002, 1005, 'Rejected');
```

-- Insert into INVENTORY table

```
INSERT INTO INVENTORY (ItemID, EquipmentID, Quantity, Status)
VALUES ('INV001', 2001, 10, 'In Stock');
INSERT INTO INVENTORY (ItemID, EquipmentID, Quantity, Status)
VALUES ('INV002', 2002, 20, 'Low Stock');
INSERT INTO INVENTORY (ItemID, EquipmentID, Quantity, Status)
VALUES ('INV003', 2003, 5, 'In Stock') ;
INSERT INTO INVENTORY (ItemID, EquipmentID, Quantity, Status)
VALUES ('INV004', 2004, 7, 'Out of Stock');
INSERT INTO INVENTORY (ItemID, EquipmentID, Quantity, Status)
VALUES ('INV005', 2005, 15, 'In Stock') ;
```

-- Insert into TRAINER_SCHEDULE table

```
INSERT INTO TRAINER_SCHEDULE (ScheduleID, TrainerID, ClassID, Day, Time)
VALUES (17001, 3001, 4001, 'Monday', TO_DATE('10:00 AM', 'HH:MI AM'));
INSERT INTO TRAINER_SCHEDULE (ScheduleID, TrainerID, ClassID, Day, Time)
VALUES (17002, 3002, 4002, 'Tuesday', TO_DATE('12:00 PM', 'HH:MI AM'));
INSERT INTO TRAINER_SCHEDULE (ScheduleID, TrainerID, ClassID, Day, Time)
VALUES (17003, 3003, 4003, 'Wednesday', TO_DATE('09:00 AM', 'HH:MI AM'));
INSERT INTO TRAINER_SCHEDULE (ScheduleID, TrainerID, ClassID, Day, Time)
VALUES (17004, 3004, 4004, 'Thursday', TO_DATE('02:00 PM', 'HH:MI AM'));
INSERT INTO TRAINER_SCHEDULE (ScheduleID, TrainerID, ClassID, Day, Time)
VALUES (17005, 3005, 4005, 'Friday', TO_DATE('11:00 AM', 'HH:MI AM'));
```

3.3 Join Data

Join MEMBER and PAYMENT to Get Member Payment Details

```
SELECT
    M.MemberID,
    M.MemberName,
    P.PaymentID,
    P.AmountPaid,
    P.PaymentDate,
    P.PaymentMethod
FROM
    MEMBER M
JOIN
    PAYMENT P
ON
    M.MemberID = P.MemberID;
```

CLASSID	CLASSNAME	TRAINERNAME	EXPERTISE	SCHEDULE	MAXPARTICIPANTS
4003	Yoga	Haziq Hassan	Yoga	15-JAN-24	25
4004	Cardio	Sarah Lim	Cardio	16-JAN-24	30
4002	Pilates	Ahmad Syukri	Pilates	12-JAN-24	15
4005	Strength	Emily Wong	Strength	17-JAN-24	15
4001	Zumba	Aminah Ali	Zumba	10-JAN-24	20

Join CLASS and TRAINER to Get Class Details and Trainer Information

```
SELECT
    C.ClassID,
    C.ClassName,
    T.TrainerName,
    T.Expertise,
    C.Schedule,
    C.MaxParticipants
FROM
    CLASS C
JOIN
    TRAINER T
ON
    C.TrainerID = T.TrainerID;
```

CLASSID	CLASSNAME	TRAINERNAME	EXPERTISE	SCHEDULE	MAXPARTICIPANTS
4003	Yoga	Haziq Hassan	Yoga	15-JAN-24	25
4004	Cardio	Sarah Lim	Cardio	16-JAN-24	30
4002	Pilates	Ahmad Syukri	Pilates	12-JAN-24	15
4005	Strength	Emily Wong	Strength	17-JAN-24	15
4001	Zumba	Aminah Ali	Zumba	10-JAN-24	20

Join MEMBER, CLASS, and ATTENDANCE for Attendance Details

```
SELECT
    M.MemberID,
    M.MemberName,
    C.ClassName,
    A.AttendanceDate,
    A.AttendanceStatus
FROM
    MEMBER M
JOIN
    ATTENDANCE A
ON
    M.MemberID = A.MemberID
JOIN
    CLASS C
ON
    A.ClassID = C.ClassID ;
```

MEMBERID	MEMBERNAME	CLASSNAME	ATTENDANCEDATE	ATTENDANCESTATUS
1001	Ali Ahmad	Zumba	05-JAN-24	Present
1002	Farah Binti Ismail	Pilates	06-JAN-24	Absent
1003	John Doe	Yoga	07-JAN-24	Present
1004	Zainab Abdullah	Cardio	08-JAN-24	Present
1005	Karim Bin Hassan	Strength	09-JAN-24	Absent

Join MEMBERSHIP_PLAN and SUBSCRIPTION for Subscription Details

```
SELECT
    S.SubscriptionID,
    M.MemberID,
    M.MemberName,
    P.PlanName,
    S.StartDate,
    S.EndDate,
    S.Status
FROM
    SUBSCRIPTION S
JOIN
    MEMBER M
ON
    S.MemberID = M.MemberID
JOIN
    MEMBERSHIP_PLAN P
ON
    S.PlanID = P.PlanID;
```

SUBSCRIPTIONID	MEMBERID	MEMBERNAME	PLANNAME	STARTDATE	ENDDATE	STATUS
13003	1003	John Doe	Basic	01-JAN-24	01-FEB-24	Expired
13001	1001	Ali Ahmad	Premium	01-JAN-24	01-JAN-25	Active
13005	1005	Karim Bin Hassan	Family	15-JAN-24	15-JAN-25	Active
13002	1002	Farah Binti Ismail	Student	10-JAN-24	10-FEB-24	Expired
13004	1004	Zainab Abdullah	VIP	15-JAN-24	15-JAN-25	Active

Join TRAINER, EQUIPMENT, and TRAINER_CERTIFICATION for Trainer Equipment and Certification Details

```
SELECT
    T.TrainerID,
    T.TrainerName,
    E.EquipName,
    E.Status AS EquipmentStatus,
    C.CertificationName,
    C.IssueDate,
    C.ExpiryDate
FROM
    TRAINER T
JOIN
    EQUIPMENT E
ON
    T.TrainerID = E.TrainerID
JOIN
    TRAINER_CERTIFICATION C
ON
    T.TrainerID = C.TrainerID;
```

TRAINERID	TRAINERNAME	EQUIPNAME	EQUIPMENTSTATUS	CERTIFICATIONNAME	ISSUEDATE	EXPIRYDATE
3001	Haziq Hassan	Treadmill	Available	Yoga Instructor	01-JAN-23	01-JAN-26
3002	Sarah Lim	Dumbbells	In Use	Cardio Specialist	01-FEB-23	01-FEB-26
3003	Ahmad Syukri	Rowing Machine	Available	Pilates Trainer	05-MAR-23	05-MAR-26
3004	Emily Wong	Exercise Bike	Available	Strength Coach	01-APR-23	01-APR-26
3005	Aminah Ali	Kettlebells	Available	Zumba Instructor	01-MAY-23	01-MAY-26

Join TRAINING_SESSION, CLASS, and MEMBER for Session Feedback Details

```

SELECT
    TS.SessionID,
    M.MemberName,
    C.ClassName,
    TS.SessionDate,
    TS.Duration,
    F.FeedbackText
FROM
    TRAINING_SESSION TS
JOIN
    MEMBER M
ON
    TS.MemberID = M.MemberID
JOIN
    CLASS C
ON
    TS.ClassID = C.ClassID
JOIN
    FEEDBACK F
ON
    TS.FeedbackID = F.FeedbackID ;

```

SESSIONID	MEMBERNAME	CLASSNAME	SESSIONDATE	DURATION	FEEDBACKTEXT
15001	Ali Ahmad	Zumba	05-JAN-24	60	Great class, loved the energy!
15002	Farah Binti Ismail	Pilates	06-JAN-24	45	Would like more variations in routines.
15003	John Doe	Yoga	07-JAN-24	60	Very relaxing, felt refreshed!
15004	Zainab Abdullah	Cardio	08-JAN-24	60	Good instructor but class was too crowded.
15005	Karim Bin Hassan	Strength	09-JAN-24	45	Fun workout, would definitely come again!

Join MEMBER, HEALTH_RECORD, and WORKOUT_PLAN for Member Health and Workout Plans

```

SELECT
    M.MemberID,
    M.MemberName,
    H.Height,
    H.Weight,
    H.BMI,
    H.MedicalConditions,
    WP.PlanDetails,
    WP.Duration
FROM
    MEMBER M
JOIN
    HEALTH_RECORD H
ON
    M.MemberID = H.MemberID
JOIN
    WORKOUT_PLAN WP
ON
    M.MemberID = WP.MemberID;

```

MEMBERID	MEMBERNAME	HEIGHT	WEIGHT	BMI	MEDICALCONDITIONS	PLANEDETAILS	DURATION
1001	Ali Ahmad	170	75	25.9	None	Yoga & Cardio Routine	4
1002	Farah Binti Ismail	160	65	25.4	Hypertension	Strength & Endurance	6
1003	John Doe	180	80	24.7	Asthma	Pilates & Core Training	8
1004	Zainab Abdullah	165	55	20.2	None	Weight Loss & Toning	12
1005	Karim Bin Hassan	175	90	29.4	Diabetes	Zumba & Dance Fitness	10

Join PROMOTION and SUBSCRIPTION to See Active Promotions

```
SELECT
    PR.PromotionName,
    PR.StartDate,
    PR.EndDate,
    PR.DiscountRate,
    S.SubscriptionID,
    M.MemberName
FROM
    PROMOTION PR
JOIN
    SUBSCRIPTION S
ON
    PR.ApplicablePlans LIKE '%' || S.PlanID || '%'
JOIN
    MEMBER M
ON
    S.MemberID = M.MemberID;
```

PROMOTIONNAME	STARTDATE	ENDDATE	DISCOUNTRATE	SUBSCRIPTIONID	MEMBERNAME
New Year Special	01-JAN-24	31-JAN-24	10	13001	Ali Ahmad
Summer Fitness Deal	01-JUN-24	31-AUG-23	25	13001	Ali Ahmad
Student Discount	01-FEB-24	30-JUN-23	15	13002	Farah Binti Ismail
New Year Special	01-JAN-24	31-JAN-24	10	13003	John Doe
New Year Special	01-JAN-24	31-JAN-24	10	13004	Zainab Abdullah
Summer Fitness Deal	01-JUN-24	31-AUG-23	25	13004	Zainab Abdullah
VIP Special Member	01-APR-24	30-APR-24	30	13004	Zainab Abdullah
Family & Friends	15-JAN-24	15-FEB-24	20	13005	Karim Bin Hassan
Summer Fitness Deal	01-JUN-24	31-AUG-23	25	13005	Karim Bin Hassan

3.4 SQL Queries

Query 1: Projection on Single Column

```
SELECT MemberName
FROM MEMBER;
```

MEMBERNAME
Ali Ahmad
Farah Binti Ismail
John Doe
Zainab Abdullah
Karim Bin Hassan

Query 2: Selection Using Logical Operators (AND)

```
SELECT MemberName, MemberContactNum
FROM MEMBER
WHERE MemberAddress LIKE '%Street%' AND MemberContactNum LIKE '012%';
```

MEMBERNAME	MEMBERCONTACTNUM
Ali Ahmad	012-3456789

Query 3: Selection Using Comparison Operator (Using = to Find Exact Matches)

```
SELECT MemberID, MemberName, MemberAddress
FROM MEMBER
WHERE MemberAddress = '123 Green Street';
```

MEMBERID	MEMBERNAME	MEMBERADDRESS
1001	Ali Ahmad	123 Green Street

Query 4: Selection Using Comparison Operator (Using < to Filter by Numeric Values)

```
SELECT ClassID, ClassName, MaxParticipants
FROM CLASS
WHERE MaxParticipants < 20 ;
```

CLASSID	CLASSNAME	MAXPARTICIPANTS
4002	Pilates	15
4005	Strength	15

Query 5: Pattern Matching using LIKE : Specified Word

```
SELECT FacilityName
FROM FACILITY
WHERE FacilityName LIKE '%Gym%';
```

FACILITYNAME
Gym Floor

Query 6: Pattern Matching using LIKE : The First Character is C

```
SELECT TrainerName
FROM TRAINER
WHERE Expertise LIKE 'C%' ;
```

TRAINERNAME
Sarah Lim

Query 7: Aggregate Function: Calculating sum using SUM()

```
SELECT MemberID, SUM(AmountPaid) AS TotalPayment
FROM PAYMENT
GROUP BY MemberID;
```

MEMBERID	TOTALPAYMENT
1004	150
1001	1200
1002	100
1005	600
1003	1500

Query 8: Aggregate Function: Counting Total Members using (COUNT)

```
SELECT COUNT(*) AS TotalMembers
FROM MEMBER;
```

TOTALMEMBERS
5

Query 9: Join between Two Tables

```
SELECT M.MemberName, C.ClassName
FROM MEMBER M
JOIN PAYMENT P ON M.MemberID = P.MemberID
JOIN CLASS C ON P.ClassID = C.ClassID;
```

MEMBERNAME	CLASSNAME
Ali Ahmad	Zumba
Farah Binti Ismail	Pilates
John Doe	Yoga
Zainab Abdullah	Cardio
Karim Bin Hassan	Strength

Query 10: Query member that join subscription and membership plan (JOIN)

```
SELECT MemberName, PlanName, StartDate, EndDate
FROM MEMBER
JOIN SUBSCRIPTION ON MEMBER.MemberID = SUBSCRIPTION.MemberID
JOIN MEMBERSHIP_PLAN ON SUBSCRIPTION.PlanID = MEMBERSHIP_PLAN.PlanID
WHERE EndDate < SYSDATE;
```

MEMBERNAME	PLANNAME	STARTDATE	ENDDATE
Farah Binti Ismail	Student	10-JAN-24	10-FEB-24
John Doe	Basic	01-JAN-24	01-FEB-24

Query 11: Query all members, even those without payments (LEFT JOIN)

```
SELECT M.MemberName, P.AmountPaid
FROM MEMBER M
LEFT JOIN PAYMENT P ON M.MemberID = P.MemberID
```

MEMBERNAME	AMOUNTPAID
Ali Ahmad	1200
Farah Binti Ismail	100
John Doe	1500
Zainab Abdullah	150
Karim Bin Hassan	600

Query 12: Using DISTINCT to find unique member contact numbers

```
SELECT DISTINCT MemberContactNum  
FROM MEMBER;
```

MEMBERCONTACTNUM
012-3456789
017-2349876
019-8765432
013-5674321
016-9834567

Query 13: Union Query to get all unique classes and trainers

```
SELECT ClassName FROM CLASS  
UNION  
SELECT Expertise FROM TRAINER;
```

CLASSNAME
Cardio
Pilates
Strength
Yoga
Zumba

Query 14: Sorting Members by Name in Ascending Order (ASC)

```
SELECT MemberID, MemberName, MemberAddress
FROM MEMBER
ORDER BY MemberName ASC;
```

MEMBERID	MEMBERNAME	MEMBERADDRESS
1001	Ali Ahmad	123 Green Street
1002	Farah Binti Ismail	456 Blue Avenue
1003	John Doe	789 Red Road
1005	Karim Bin Hassan	654 Orange Crescent
1004	Zainab Abdullah	321 Yellow Lane

Query 15: Sorting Members by Total Payment (Using JOIN) and Descending order (DESC)

```
SELECT MEMBER.MemberID, MemberName, SUM(PAYMENT.AmountPaid) AS TotalPaid
FROM MEMBER
JOIN PAYMENT ON MEMBER.MemberID = PAYMENT.MemberID
GROUP BY MEMBER.MemberID, MemberName
ORDER BY TotalPaid DESC;
```

MEMBERID	MEMBERNAME	TOTALPAID
1003	John Doe	1500
1001	Ali Ahmad	1200
1005	Karim Bin Hassan	600
1004	Zainab Abdullah	150
1002	Farah Binti Ismail	100

3.5 Update and Delete Queries

Updating A Single Column

```
UPDATE MEMBER  
SET MemberAddress = 'Penang'  
WHERE MemberID = 1002;
```

MEMBERID	MEMBERNAME	MEMBERADDRESS
1001	Ali Ahmad	123 Green Street
1002	Farah Binti Ismail	Penang
1003	John Doe	789 Red Road
1004	Zainab Abdullah	321 Yellow Lane
1005	Karim Bin Hassan	654 Orange Crescent

Updating Multiple Column

```
UPDATE TRAINER  
SET TrainerName = 'Izzati Cantik', Expertise = 'Strength oo'  
WHERE TrainerID = 3004;
```

TRAINERID	TRAINERNAME	EXPERTISE
3001	Haziq Hassan	Yoga
3002	Sarah Lim	Cardio
3003	Ahmad Syukri	Pilates
3004	Izzati Cantik	Strength oo
3005	Aminah Ali	Zumba

Using Comparison Operators

```
UPDATE CLASS
SET MaxParticipants = 30
WHERE MaxParticipants < 20 ;
```

CLASSID	CLASSNAME	MAXPARTICIPANTS
4001	Zumba	20
4002	Pilates	30
4003	Yoga	25
4004	Cardio	30
4005	Strength	30

Using a Subquery in UPDATE

```
UPDATE PAYMENT
SET AmountPaid = (SELECT MAX(AmountPaid) FROM PAYMENT)
WHERE MemberID = 1001;
```

PAYMENTID	MEMBERID	AMOUNTPAID
5001	1001	1500
5002	1002	100
5003	1003	1500
5004	1004	150
5005	1005	600

Update based on Date Condition

```
UPDATE PAYMENT
SET PaymentMethod = 'Online Transfer'
WHERE PaymentDate < TO_DATE('08/01/2024', 'MM/DD/YYYY') AND PaymentMethod = 'Cash';
```

PAYMENTID	MEMBERID	AMOUNTPAID	PAYMENTDATE	PAYMENTMETHOD	CLASSID
5001	1001	1500	05-JAN-24	Credit Card	4001
5002	1002	100	06-JAN-24	Online Transfer	4002
5003	1003	1500	07-JAN-24	Debit Card	4003
5004	1004	150	08-JAN-24	Online Transfer	4004
5005	1005	600	09-JAN-24	Credit Card	4005

Delete a member and all related record for certain member

```
DELETE FROM MEMBER
WHERE MemberID = 1001;
```

```
SELECT *
FROM MEMBER;
```

MEMBERID	MEMBERNAME	MEMBERADDRESS	MEMBERCONTACTNUM
1002	Farah Binti Ismail	Penang	019-8765432
1003	John Doe	789 Red Road	013-5674321
1004	Zainab Abdullah	321 Yellow Lane	017-2349876
1005	Karim Bin Hassan	654 Orange Crescent	016-9834567

Delete selected row

```
DELETE FROM FEEDBACK
WHERE FEEDBACKID = 12002;
```

```
SELECT *
FROM FEEDBACK;
```

FEEDBACKID	MEMBERID	CLASSID	TRAINERID	FEEDBACKTEXT
12003	1003	4003	3001	Very relaxing, felt refreshed!
12004	1004	4004	3002	Good instructor but class was too crowded.
12005	1005	4005	3003	Fun workout, would definitely come again!

Delete Members Who Have Not Made Payments Greater Than a Certain Amount

```
DELETE FROM MEMBER
WHERE MemberID IN (
    SELECT M.MemberID
    FROM MEMBER M
    JOIN PAYMENT P ON M.MemberID = P.MemberID
    GROUP BY M.MemberID
    HAVING SUM(P.AmountPaid) > 1000
)
```

```
SELECT * FROM MEMBER;
```

MEMBERID	MEMBERNAME	MEMBERADDRESS	MEMBERCONTACTNUM
1002	Farah Binti Ismail	Penang	019-8765432
1004	Zainab Abdullah	321 Yellow Lane	017-2349876
1005	Karim Bin Hassan	654 Orange Crescent	016-9834567

4.0 Conclusion

The Fitness Club Membership System project has provided a thorough solution for handling the various aspects of fitness club operation. The project successfully centralized the management of trainer assignments, class schedules, member information, payment records, equipment maintenance by implementing the relational database. Besides that, reducing errors and redundancies that come with manual data handling, this approach has improved the club's ability to deliver a smooth and personalized experience to its each member.

One significant achievement of this project is the database, including scheduling maintenance, managing membership, keeping track of class attendance, and generating financial reports. These automated features reducing a lot of the management burden, enabling staff to focus more on member engagement and service quality. Furthermore, all members sensitive data, including health records and payment information is protected and accessible only those with permission can access control to the system's base. Compliance with data regulations laws strengthen the member trust in the club dedication to privacy.

To provide flexibility, the project's entity relationship design accurately represents the different links between participants, instructors, classes, equipment, and facilities. the system also has comprehensive modules like the data dictionary, ERD, and SQL queries, which provide club management the ability to efficiently track outcome. These features enable the managers to make data-driven decisions making and strategic planning by tracking peak usage time, analysing attendance trends, gain insights into revenue and membership development in details.

In conclusion, by offering a dependable, safe, and efficient database management system, the Fitness Club Membership System project has successfully met its goals. In addition to enhancing member satisfaction and operational efficiency, this system organizes the fitness centre for any potential future growth. The system's scalability ensures it to support more locations, add new classes and handle expanding member base ensuring it will help the club towards long-term success in a highly competitive sector.

5.0 APPENDIX

LINK ORACLE LIVE SQL:

<https://livesql.oracle.com/ords/livesql/s/cl6driti7n5w6jxes9jcqg63n>